

SYNTHETIC DATA FOR MACHINE TRANSLATION

Where we are: back-translation, fuzzy matches, generated text and its features

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WHY MACHINE TRANSLATION, AND WHY SPECIALISED DOMAINS

Stage 1 / 3



**The aim: make text
accessible across languages**

Far more text than professional translators can handle. Automation is needed.



WHY MACHINE TRANSLATION, AND WHY SPECIALISED DOMAINS

Stage 2 / 3



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General domain: good enough for most uses

News, everyday text, little specialised terminology.

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Stage 3 / 3



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Specialised domains: not there yet

Legal, biomedical, technical. Terminology, conventions, formulaic style.

Why not?



THE DATA PROBLEM IN SPECIALISED TRANSLATION

Stage 1 / 3

ONE SENTENCE PAIR FROM EU LEGISLATION

EN Debt, breakdown by residual maturity



FR Dette, ventilation par échéance
résiduelle

DGT translation memory, English to French



THE DATA PROBLEM IN SPECIALISED TRANSLATION

Stage 2 / 3

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DGT translation memory, English to French



Translation models learn from sentence pairs

A source sentence and its translation.



Specialised domains have few pairs

300,000 is small. Most domains have fewer.



Monolingual text is easier to find

Use it to create the missing half.



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Use it to create the missing half.

Three ways to create the missing half

1 Translate real text backwards

2 Add retrieval

3 Generate the text



METHOD 1: BACK-TRANSLATION

Stage 1 / 2



The English source is missing

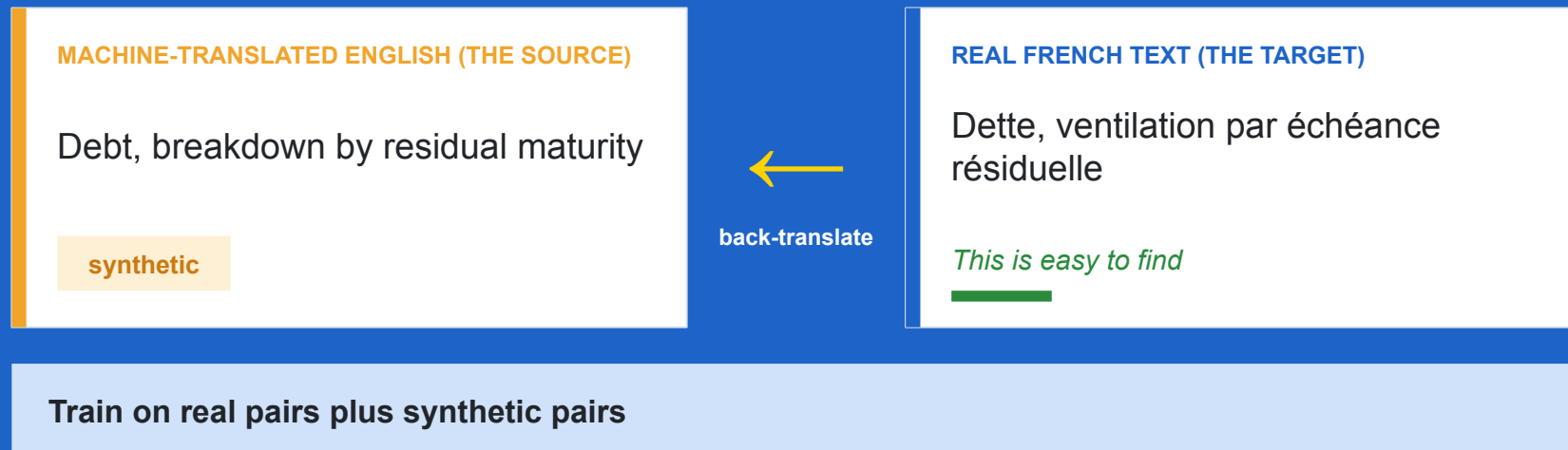
REAL FRENCH TEXT (THE TARGET)

Dette, ventilation par échéance
résiduelle

This is easy to find

METHOD 1: BACK-TRANSLATION

Stage 2 / 2



WHAT IS KNOWN ABOUT IT

- 1 The gain comes from the real target side
- 2 A tag or some noise marks what is synthetic
- 3 More synthetic data stops helping at some point

[Sennrich et al. 2016]
[Edunov et al. 2018]
[Caswell et al. 2019]
[Poncelas et al. 2018]



METHOD 2: RETRIEVAL AND BACK-TRANSLATION

Stage 1 / 3

WHAT A FUZZY MATCH IS

Input (English)

Debt, breakdown by residual maturity

Closest sentence in the memory, similarity 0.98

Debt, breakdown by initial maturity

[Bulte & Tezcan 2019]

METHOD 2: RETRIEVAL AND BACK-TRANSLATION

Stage 2 / 3

WHAT A FUZZY MATCH IS

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Debt, breakdown by initial maturity

Its stored translation, given to the model

Dette, ventilation par échéance initiale

Output

Dette, ventilation par échéance résiduelle

[Bulte & Tezcan 2019]

METHOD 2: RETRIEVAL AND BACK-TRANSLATION

Stage 3 / 3

WHAT A FUZZY MATCH IS

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Dette, ventilation par échéance initiale

Output

Dette, ventilation par échéance résiduelle

+12.10 BLEU

over the best baseline, English to Ukrainian

64 to 86%

of the retrieved matches come from back-translated sentences

Back-translated pairs enlarge the memory.

[Bulte & Tezcan 2019]

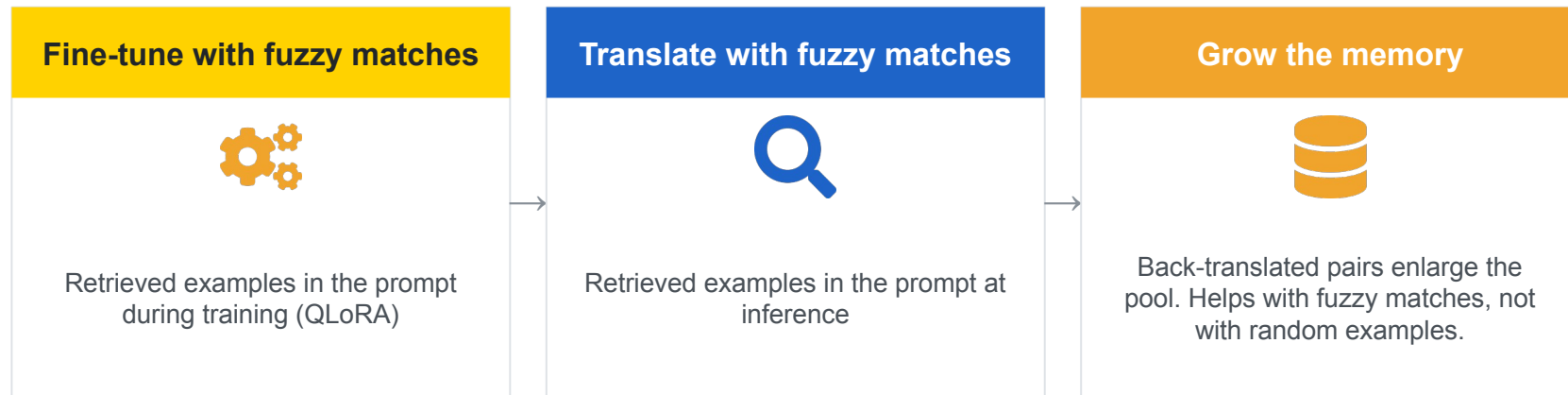
[Moerman & Tezcan 2024]

[Tezcan, Skidanova & Moerman 2024]



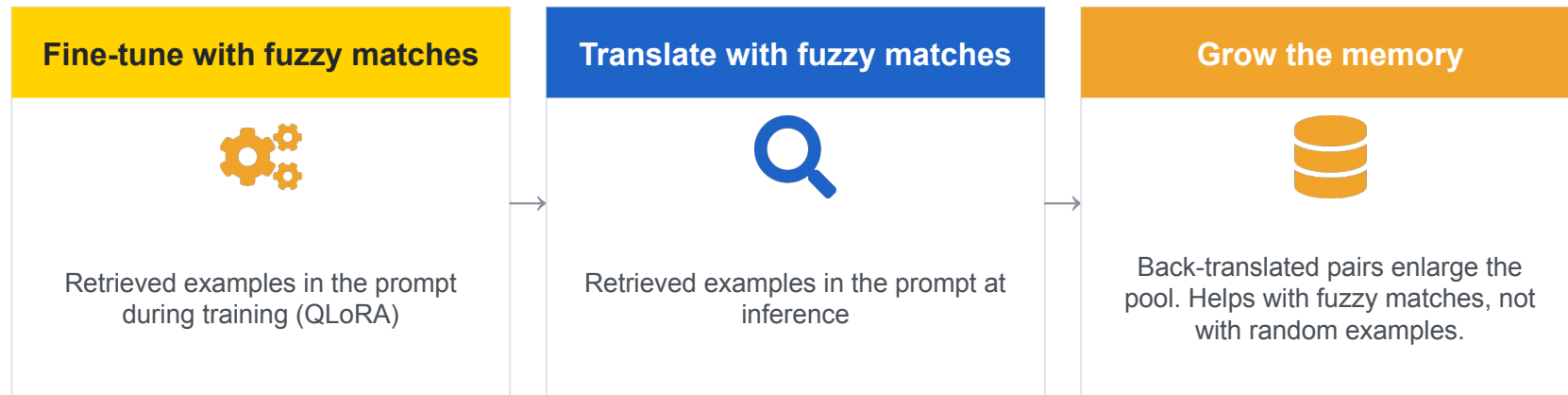
THE SAME APPROACH WITH LLMS

Stage 1 / 2



THE SAME APPROACH WITH LLMS

Stage 2 / 2



+10.48 COMET · +10.73 BLEU

over neural MT in lower-resource settings

[Moslem et al. 2023]

[Zhang et al. 2023]

[Moerman, Lefever & Tezcan, under review]



APPLIED TO A SHARED TASK: UPPER AND LOWER SORBIAN

Stage 1 / 3

WMT 2026, Sorbian track: one 2B model for five tasks, and almost no task-specific training data.

THREE KINDS OF CREATED DATA



Back-translated pairs, used for retrieval only

665,738 synthetic German to Sorbian pairs, used only in the retrieval memory. Kept out of fine-tuning: +6.26 chrF++.



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Task data made from dictionaries and monolingual text

Typos for spell checking, paradigm swaps for grammar checking, cloze items for QA. 100k items where the organisers had 4,000.



Seed problems translated by our own MT engine

28,147 GSM8K and MATH problems translated into Sorbian by our own MT engine. The organisers had 48.



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25.8 → 76.1

MT chrF++ on the dev sets: base model with fuzzy matches, then fine-tuned with fuzzy matches

Also with a 358-sentence memory and 14 target languages: fuzzy matches beat random examples for 13 of 14 (EAMT 2026).



METHOD 3: TEXT GENERATION

Stage 1 / 3

A language model writes the target text. Back-translation turns it into pairs.

Written by Mistral-7B after a few DGT examples:

“L'article 201 quinquès est modifié pour devenir le texte ci-après”

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2022 · Moslem et al.

LLM writes in-domain text, then back-translation: +5 to 6 BLEU with almost no data.

2024 · NewsPaLM (Finkelstein et al.)

LLM-generated pairs beat a crawled corpus 300 times larger.

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Noisy GPT-4o translations still help low-resource MT: +3 to +21 chrF.

2026 · TranslateGemma

Trained on a synthetic-plus-human mix as standard practice.



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Generated text helps. Which generated text helps most is the open question.



OUR STUDY: WHICH GENERATED TEXT HELPS?

Stage 1 / 2



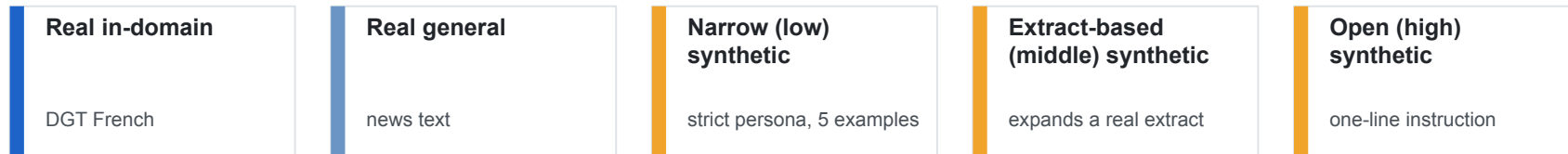
[Moerman et al., in preparation]

OUR STUDY: WHICH GENERATED TEXT HELPS?

Stage 2 / 2



WHAT VARIES: THE POOL



Two pool sizes: 50k and 200k sentences.

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[Moerman et al., in preparation]



WHAT WE MEASURED ON EVERY POOL

Stage 1 / 4

1 Diversity

2 Domain proximity

3 Relation to the test set

4 Naturalness and shape

How varied is the text?

Lexical diversity

How many different words does the pool use?

EXAMPLE

Low: every sentence starts with "Les États membres veillent à ce que...". High: the same idea in different words.

Syntactic diversity

How many different sentence structures?

EXAMPLE

"L'article 5 est remplacé par le texte suivant." vs "Le texte suivant remplace l'article 5." vs "À l'article 5, le texte est remplacé comme suit."

Semantic diversity

How many different topics?

EXAMPLE

Low: only fisheries quotas. High: fisheries, customs, chemicals, sanctions, aid.



WHAT WE MEASURED ON EVERY POOL

Stage 2 / 4

1 Diversity

2 Domain proximity

3 Relation to the test set

4 Naturalness and shape

Does it look like EU legislation?

Closeness in meaning space

Does the pool sit near the legal corpus when both are embedded?

EXAMPLE

Near: "Debt, breakdown by residual maturity". Far: "The striker scored twice".

Surprise for a legal LM

Is a model trained on legislation surprised by the sentence?

EXAMPLE

Regulation clause: not surprised. Football report: very. Generated legal text: least of all.

Topic coverage

Does the pool fall into the known legal topics?

EXAMPLE

DGT topics from clustering: fisheries, customs tariffs, REACH, sanctions, development aid.



WHAT WE MEASURED ON EVERY POOL

Stage 3 / 4

1 Diversity

2 Domain proximity

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4 Naturalness and shape

Does it cover what we will be asked to translate?

Retrieval coverage

For each test sentence, is there a near neighbour in the pool?

EXAMPLE

Test: "... residual maturity". Nearest in the pool: "... initial maturity" (0.98, real example).

Phrase overlap

Do the test set's word sequences occur in the pool?

EXAMPLE

"breakdown by", "residual maturity": in the pool or not.

Rare-term coverage

Are the rare legal terms of the test set present?

EXAMPLE

"échéance résiduelle": rare in general French, frequent in DGT.



WHAT WE MEASURED ON EVERY POOL

Stage 4 / 4

1 Diversity

2 Domain proximity

3 Relation to the test set

4 Naturalness and shape

Does it read and look like human text?

Fluency for a general LM

Does an ordinary French language model find the sentence natural?

EXAMPLE

Generated legal French reads smoothly. Real legal French is terse and full of references.

Shape of the corpus

Do sentence lengths and word frequencies look like real text?

EXAMPLE

Generated: 15 to 20 words per sentence. Real: 3 to 60, with a long tail of rare words.

The back-translation round trip

What does the back-translator do with the text?

EXAMPLE

Length and vocabulary of the back-translated English, and the filter drop rate.

[Guo et al. 2025]

[Moerman et al., in preparation]



HOW THE SYNTHETIC POOLS WERE MADE

Stage 1 / 2

Same generator, same real seeds, same cleaning. Only the prompt differs.

Then: split into sentences, deduplicate, remove overlap with the training data, keep French, take the first 50k or 200k.

[Moerman et al., in preparation]

HOW THE SYNTHETIC POOLS WERE MADE

Stage 2 / 2

Same generator, same real seeds, same cleaning. Only the prompt differs.

Narrow (low)

least varied, most legal-looking

PERSONA

"A distinguished French legal scholar and practitioner ..."

EXAMPLES

5 real DGT sentences as reference

ASKS FOR

3 sentences in the same professional style, numbered

Extract-based (middle)

in between on every measure

PERSONA

"A senior legal linguist for the European Commission"

EXAMPLES

One 50-word DGT extract, plus a topic, an audience and a format

ASKS FOR

A text that expands the extract

Open (high)

most varied, least legal-looking

PERSONA

"A legal writing assistant"

EXAMPLES

5 real DGT sentences

ASKS FOR

"3 legal sentences similar to these. Vary the syntax and structure."

Then: split into sentences, deduplicate, remove overlap with the training data, keep French, take the first 50k or 200k.

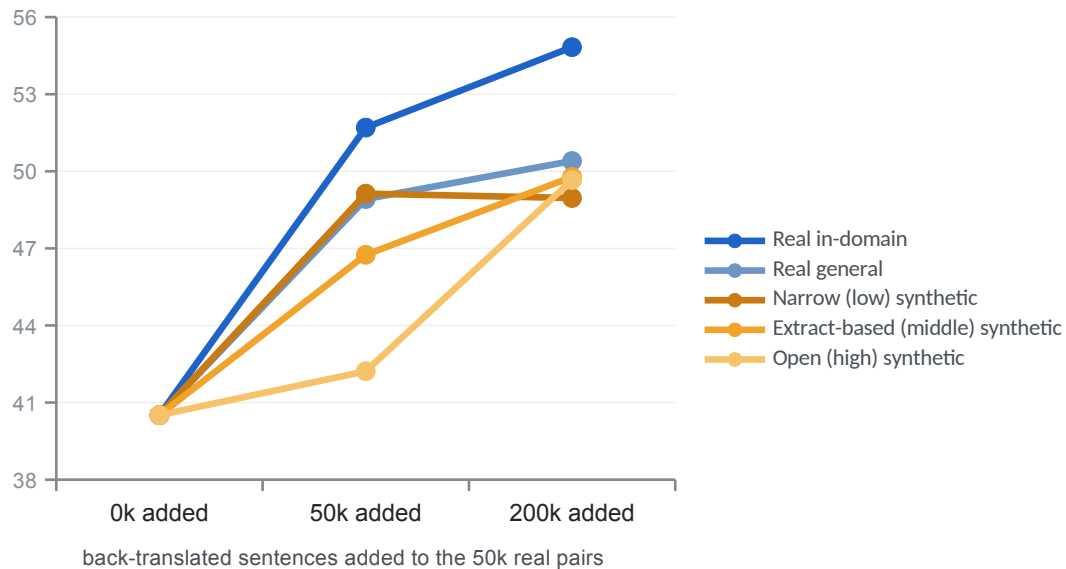
[Moerman et al., in preparation]



RESULTS BY POOL AND POOL SIZE

Stage 1 / 2

BLEU on the DGT test set, Mistral-7B, retrieved examples at inference for every point



54.83

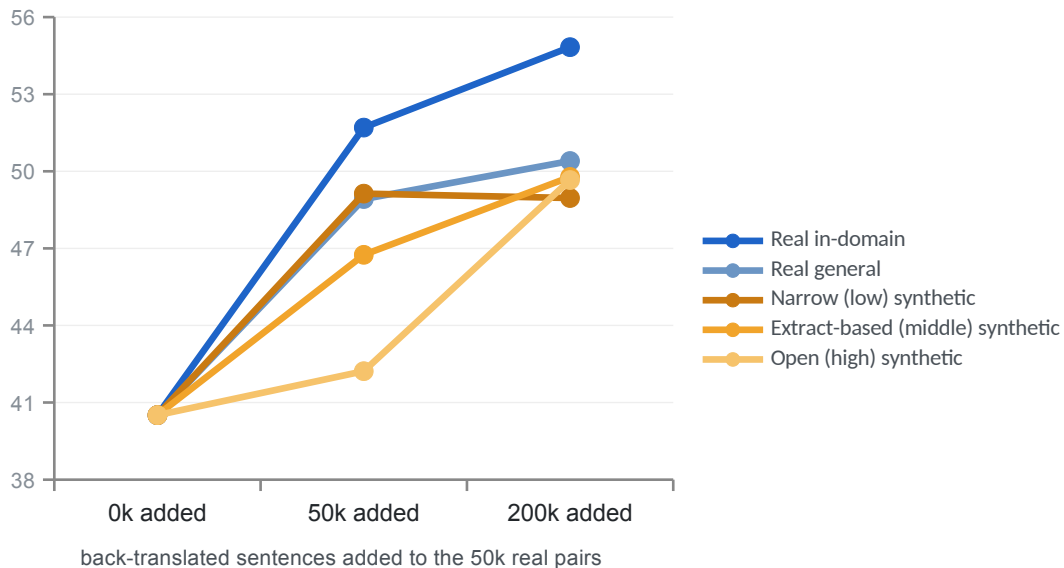
real in-domain with 200k added



RESULTS BY POOL AND POOL SIZE

Stage 2 / 2

BLEU on the DGT test set, Mistral-7B, retrieved examples at inference for every point



54.83

real in-domain with 200k added

49.78

best synthetic with 200k added

40.51

start of every line: no added data, no fine-tuning. Fine-tuned on the real pairs alone: 48.69.



DO THE MEASUREMENTS PREDICT THE RESULTS?

Stage 1 / 4

Measures on the 50k pools, and the translation scores. Pools ordered by prompt.

Pool	BLEU 50k
Narrow (low) synthetic	49.1
Extract-based (middle) synthetic	46.7
Open (high) synthetic	42.2
Real in-domain	51.7

1 / 4

At 50k the pools differ a lot. Narrow is best, open is 7 BLEU behind.



DO THE MEASUREMENTS PREDICT THE RESULTS?

Stage 2 / 4

Measures on the 50k pools, and the translation scores. Pools ordered by prompt.

Pool	Lexical diversity	Surprise, legal LM	Test-set coverage	Surprise, general LM	BLEU 50k
<i>how to read</i>	<i>higher: more varied</i>	<i>higher: less legal-like</i>	<i>higher: closer to test set</i>	<i>higher: less everyday French</i>	<i>higher: better</i>
Narrow (low) synthetic	0.35	5.4	0.59	4.4	49.1
Extract-based (middle) synthetic	0.41	7.1	0.66	6.9	46.7
Open (high) synthetic	0.49	9.7	0.67	9.3	42.2
Real in-domain	0.53	16.2	0.70	19.5	51.7

Every measure increases from top to bottom, in the order of the prompts.

2 / 4

Every measure orders the pools like the prompts do: narrow, extract-based, open. The measures restate the prompt.



DO THE MEASUREMENTS PREDICT THE RESULTS?

Stage 3 / 4

Measures on the 200k pools (coverage and fluency were measured at 50k only), and the translation scores.

Pool	Lexical diversity	Surprise, legal LM	Test-set coverage	Surprise, general LM	BLEU 50k	BLEU 200k
<i>how to read</i>	<i>higher: more varied</i>	<i>higher: less legal-like</i>	<i>higher: closer to test set</i>	<i>higher: less everyday French</i>	<i>higher: better</i>	<i>higher: better</i>
Narrow (low) synthetic	0.40	6.6	0.59	4.4	49.1	49.0
Extract-based (middle) synthetic	0.47	9.3	0.66	6.9	46.7	49.8
Open (high) synthetic	0.52	11.8	0.67	9.3	42.2	49.7
Real in-domain	0.53	16.2	0.70	19.5	51.7	54.8

3 / 4

At 200k the synthetic pools are within one BLEU of each other. The measures still differ.



DO THE MEASUREMENTS PREDICT THE RESULTS?

Stage 4 / 4

Measures on the 200k pools (coverage and fluency were measured at 50k only), and the translation scores.

Pool	Lexical diversity	Surprise, legal LM	Test-set coverage	Surprise, general LM	BLEU 50k	BLEU 200k
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4 / 4

Real text is ahead at both sizes. By these measures it looks least like legislation and least fluent.



RELATED FINDINGS AND NEXT STEPS

Stage 1 / 2

THE FIELD AGREES

Mix synthetic with real data

One third synthetic sped up pretraining 5 to 10 times. Synthetic alone did not.

[Kang et al. 2025]

Generated text is less diverse than human text

Lexically, syntactically, semantically. Training on it repeatedly narrows it further.

[Guo et al. 2025, Shumailov et al. 2024]

No metric replaces training and evaluating

90 experiments, over a trillion tokens. The best proxy explained 7% of the variance.

[Niklaus et al. 2026]



RELATED FINDINGS AND NEXT STEPS

Stage 2 / 2

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WHAT WE DO NEXT



A practical guide for in-domain MT data

What to generate, what to select, what not to spend compute on.



Does synthetic data need to be human-like?

Make generated text harder to tell from human text. Does that close the gap, or does human-likeness not matter?

[García-Romero et al. 2025, Sánchez-Cartagena et al. 2024]



The same method beyond MT

Contrastive synthetic sentences teach a 4B model to detect gender ambiguity.

[Moerman, Gkovedarou & Hackenbuchner 2026]



THANK YOU

In short: back-translation still works, retrieval makes it work better, and we do not yet know what makes real text more useful than generated text.

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REFERENCES

OWN WORK

Moerman T., Tezcan A. (2024). Leveraging synthetic monolingual data for fuzzy-match augmentation in NMT. KEMT workshop, EAMT.

Tezcan A., Skidanova A., Moerman T. (2024). Improving fuzzy match augmented NMT in specialised domains through synthetic data. PBML 122.

Moerman T., Vanallemeersch T., Szoc S., Tezcan A. (2025). Tailoring MT for scientific literature through topic filtering and fuzzy match augmentation. PSLT.

Moerman T., Lefever E., Tezcan A. Advancing fuzzy match augmentation for domain-specific MT: LLMs and NMT. JAIR, under review.

Moerman T., Tezcan A., Macken L. (2026). Multilingual communication in the asylum context. EAMT.

Moerman T., Tezcan A. (2026). Getting more from small LLMs: fuzzy-match retrieval and candidate scoring for multitask Sorbian NLP. WMT.

Moerman T., Gkovedarou E., Hackenbuchner J. (2026). ContraGAND. EMNLP.

BACK-TRANSLATION

Sennrich R., Haddow B., Birch A. (2016). Improving NMT models with monolingual data. ACL.

Edunov S. et al. (2018). Understanding back-translation at scale. EMNLP.

Poncelas A. et al. (2018). Investigating backtranslation in NMT. EAMT.

Caswell I., Chelba C., Grangier D. (2019). Tagged back-translation. WMT.

Marie B., Rubino R., Fujita A. (2020). Tagged back-translation revisited. ACL.

Edunov S. et al. (2020). On the evaluation of MT systems trained with back-translation. ACL.

RETRIEVAL

Bulte B., Tezcan A. (2019). Neural fuzzy repair. ACL.

Xu J., Crego J., Senellart J. (2020). Boosting NMT with similar translations. ACL.

Cai D. et al. (2021). NMT with monolingual translation memory. ACL.

Moslem Y., Haque R., Kelleher J., Way A. (2023). Adaptive MT with large language models. EAMT.

Zhang X., Rajabi N., Duh K., Koehn P. (2023). MT with LLMs: prompting, few-shot learning, and fine-tuning with QLoRA. WMT.

Bouthors M., Crego J., Yvon F. (2024). Retrieving examples from memory for retrieval augmented NMT. Findings of NAACL.

Zebaze A., Sagot B., Bawden R. (2025). In-context example selection via similarity search improves low-resource MT. Findings of NAACL.

LLM GENERATION AND DIVERSITY

Moslem Y., Haque R., Way A. (2022). Domain-specific text generation for MT. AMTA.

Xu H. et al. (2024). A paradigm shift in MT: boosting translation performance of LLMs (ALMA). ICLR.

Finkelstein M. et al. (2024). NewsPaLM MBR and QE dataset. WMT.

de Gibert O. et al. (2025). Scaling low-resource MT via synthetic data generation with LLMs. EMNLP.

Finkelstein M. et al. (2026). TranslateGemma technical report. arXiv 2601.09012.

Guo Y. et al. (2025). Benchmarking linguistic diversity of LLMs. TACL.

Shumailov I. et al. (2024). AI models collapse when trained on recursively generated data. Nature.

Kang F. et al. (2025). Demystifying synthetic data in LLM pre-training. EMNLP.

Niklaus J. et al. (2026). The Synthetic Data Playbook (FinePhrase). Hugging Face.

García-Romero C., Esplà-Gomis M., Sánchez-Martínez F. (2025). Automatic machine translation detection using a surrogate multilingual translation model. To appear in TACL.

Sánchez-Cartagena V. M., Esplà-Gomis M., Pérez-Ortiz J. A., Sánchez-Martínez F. (2024). Non-fluent synthetic target-language data improve neural machine translation. IEEE TPAMI 46(2).



FULL ARM TABLE, FINE-TUNED MODELS

Pool	Kind	BLEU 50k	BLEU 200k	ChrF 200k	COMET 200k
No added data	anchor	48.69		69.20	0.8631
Real in-domain (DGT)	real	51.70	54.83	73.57	0.8750
Real general (news)	real	48.92	50.40	70.40	0.8661
Open (high) synthetic, prompt v2	synthetic	42.22	49.66	70.02	0.8670
Extract-based (middle) synthetic, prompt v7	synthetic	46.75	49.78	70.14	0.8659
Narrow (low) synthetic, prompt v4	synthetic	49.13	48.96	69.50	0.8635
50/50 news + extract-based synthetic	combination	48.26	49.90	70.16	0.8661
Three synthetic tiers mixed	combination	48.57	49.78	70.12	0.8676

Headline cell with retrieved examples at inference. The no-added-data anchor has ChrF 69.20 and COMET 0.8631 at its single size. Noise floor from a seed retrain: 0.66 BLEU, 0.0019 COMET.

Source: p1_arm_table_full.csv, FT arm

WMT 2026 SORBIAN TRACK, OFFICIAL RESULTS

Team	MT chrF++	QA acc.	SC	GC	MR acc.	Points
LT3 (ours, primary)	69.57	65.05	76.11	78.32	30.40	22
HeyBusan	72.75	55.92	78.03	79.52	29.00	22
Hitsz	63.12	57.33	73.13	62.11	18.60	16
HSE	61.50	46.77	63.63	50.02	5.00	9
Organiser baseline	21.68	43.39	6.66	1.34	5.40	5

SC and GC are the mean of detection and correction exact match. Points are the organisers' rank-based aggregate. Our system ranks first on QA and MR, second on MT, SC and GC. Development-set ablation (Table 3): fuzzy carriers in fine-tuning +3.63 chrF++ over random, authentic-only fine-tuning rows +6.26 over authentic plus back-translated rows, back-translated pairs in the retrieval pool at inference +0.06.

Source: WMT 2026 paper Table 10, official hidden-test results, Sorbian track

WHY THE METRICS CANNOT RANK THE SYNTHETIC SOURCES

Metric	rho with prompt template	p	template proxy?
semantic diversity	-0.94	0.001	yes
DGT-LM perplexity	-0.94	0.001	yes
general-LM fluency	-0.94	0.001	yes
back-translation TTR	-0.94	0.001	yes
lexical diversity	-0.87	0.011	yes
embedding distance to DGT	+0.87	0.011	yes
syntactic diversity	-0.76	0.049	yes
test-set coverage (retrieval)	-0.76	0.049	yes
validation coverage	-0.76	0.049	yes
cluster coverage	+0.53	0.222	no

READING

Spearman rho over the 7 synthetic arms between each metric and the prompt ordinal (open = 1, extract-based = 2, narrow = 3). The metrics measure the prompt, not the value of the data. COMET tells the same story as BLEU: real in-domain leads by 0.0080 at 200k, four times the COMET noise floor.

Source: p1_template_collinearity.csv

PER-LANGUAGE CHRF, 14 LANGUAGES, 4 SYSTEMS

Language	Tier	MMT+ (deployed)	MMT	TG-27B + FM	Gemini Pro + FM
Arabic	High	66.76	63.78	63.63	63.76
German	High	73.67	69.26	70.97	73.85
Spanish	High	72.64	69.77	69.98	74.30
Farsi	High	54.01	51.36	58.43	55.03
Portuguese	High	80.97	75.48	75.65	76.54
Romanian	High	67.97	63.10	63.80	67.22
Russian	High	67.62	65.25	67.95	67.01
Turkish	High	65.34	62.20	65.47	65.60
Armenian	Mid	74.26	68.58	68.61	73.42
Albanian	Mid	70.01	65.46	61.15	70.05
Georgian	Mid	79.87	83.74	61.70	65.69
Pashto	Low	66.82	64.86	57.10	62.53
Somali	Low	58.19	55.57	54.47	57.34
Tigrinya	Low	34.29	29.48	29.99	43.00
Average		66.60	63.42	62.06	65.38

Source: EAMT 2026 Table 4. Both LLMs use 15 fuzzy-match examples. MMT+ is ModernMT with the 358-sentence memory.



HOW EACH PROPERTY IS MEASURED

Property	How it is measured	Sample
Lexical diversity	unique 1-, 2-, 3-gram ratio, averaged	40k tokens
Syntactic diversity	Weisfeiler-Lehman kernel over Stanza dependency trees, 1 minus mean similarity	2,000 sentences
Semantic diversity	1 minus mean pairwise cosine of all-mpnet-base-v2 embeddings, rescaled to 0 to 1	2,000 sentences
Closeness in meaning space	mean top-10 mpnet cosine to the DGT training corpus	2,000 vs 10k
Surprise for a legal LM	per-sentence perplexity under the DGT-fine-tuned Mistral-7B back-translator	2,000 sentences
Topic coverage	share with max cosine to a DGT cluster centroid at or above 0.5	2,000 sentences
Retrieval coverage	mean top-5 mpnet cosine from test (or validation) sentences to the pool	2,000 vs 50k
Phrase overlap	share of test 1-, 2-, 3-gram types found in the pool	full
Rare-term coverage	share of 5,000 test bigrams with training count of 2 or less found in the pool	full
Fluency for a general LM	per-sentence perplexity under Mistral-7B-v0.3 base	2,000 sentences
Shape of the corpus	JS divergence of length histogram vs real, Zipf slope, rare-word rate, repeated-bigram share	50k
Back-translation round trip	EN/FR token ratio, EN type-token ratio, filter drop rate	full BT output

Every pool was sampled with the same seed. Sampled metrics are size-invariant by construction, so a full-pool unique-n-gram ratio was added for the 50k vs 200k comparison. Correlations: Spearman, computed separately for the 7 synthetic arms and for all 9 arms.

Source: `measure_diversity/*.py`, Guo et al. 2025 framework



THE THREE PROMPTS IN FULL

NARROW (LOW), PROMPT V4

SYSTEM

You are a distinguished French legal scholar and practitioner with extensive experience in drafting legal documents across multiple jurisdictions. Your expertise encompasses civil law, commercial law, criminal law, administrative law, and constitutional law. You possess a deep understanding of French legal terminology, procedural requirements, and the nuanced language used in professional legal contexts. Your task is to create authentic, original French legal sentences that demonstrate proper legal syntax, appropriate formality, and accurate domain-specific vocabulary while maintaining stylistic diversity.

USER

Your mission is to compose 3 original French legal sentences that demonstrate mastery of French jurisprudential language. Each sentence should exhibit unique syntactic patterns while maintaining thematic consistency with the reference material. Draw inspiration from various legal specializations including but not limited to: contract law, tort law, corporate governance, criminal procedure, constitutional principles, administrative regulations, and international law. Ensure each generated sentence could authentically appear in professional legal documents, court decisions, or academic legal texts.

Reference examples for stylistic guidance:

[5 real DGT sentences]

Please provide 3 distinctly structured French legal sentences following the established domain and professional standards:

1.

EXTRACT-BASED (MIDDLE), PROMPT V7

SYSTEM

You are a senior legal linguist for the European Commission. Your task is to draft high-quality French legal text based on specific parameters. You must adhere to the Style Guide for French Usage in the Institutions of the European Union.

USER

Here is an extract from an EU legal document: "[50-word DGT extract]".

Write a [format] that expands upon the extract above, within the context of "[topic]".

You must write from the perspective of: [audience].

Your text should delve into the nuances of the legal topic, offering precise formulations and rigorous analysis.

Aim to:

- Inform: Provide legally accurate, well-structured content that educates the reader about the subject matter.
- Engage: Write in a tone that matches the specified persona, making complex legal concepts clear and precise.
- Illustrate: Use concrete legal references, procedural examples, or regulatory context to bring the topic to life.

Do not give a title and do not start with sentences like "Il convient de noter que" or "Il est important de", simply write the content without these introductory phrases.

Output:

OPEN (HIGH), PROMPT V2

SYSTEM

You are a legal writing assistant. Create original French legal sentences similar to the provided examples.

USER

Create 3 French legal sentences similar to these examples. Vary the syntax and structure:

[5 real DGT sentences]

New sentences: